

# From Clips to Communities: Fusing Social Video into Knowledge Graphs for Localness-Aware LLMs

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## Abstract

Understanding how people experience and assign meaning to places — what we term “localness” — is essential for designing systems that support orientation, belonging, and community integration. However, current computational approaches to modeling place often struggle to capture the multimodal, relational, and contextual richness of localness. In this paper, we introduce a novel workflow for extracting structured localness representations from geo-tagged short-form social videos using a Graph-Enhanced Retrieval-Augmented Generation (Graph-RAG) framework. Our method aligns and fuses multimodal video data from TikTok into a knowledge graph, retrieves relevant contextual evidence through a hybrid of graph-based and semantic retrieval, and then prompts large language models (LLMs) to infer fine-grained localness attributes across cognitive, relational, and physical domains. We evaluate our system using both human-annotated ground truth and qualitative analysis, demonstrating full coverage across all localness components, with especially strong performance in environmental and relational dimensions. Our findings reveal both the promises and the challenges of grounding AI systems in lived experiences of place, and we offer design implications for localness-aware technologies in digital placemaking, community informatics, and contextual computing.

## CCS Concepts

• **Human-centered computing** → **Collaborative and social computing systems and tools**; • **Computing methodologies** → **Information extraction**.

## Keywords

Localness; Knowledge graph; Large language models; Retrieval-augmented generation (RAG); Multimodal data; Place representation

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## 1 Introduction

When people move or travel to a new city or neighborhood, they often struggle to understand the everyday texture of a place [10]: Where do locals actually eat? Is that park good for quiet reading or weekend festivals? What’s the vibe of the neighborhood — family-oriented, artsy, student-heavy? While search engines and location-based services offer facts like hours, ratings, or addresses, they rarely capture the emotional, cultural, or social layers that give a place its character [18, 29, 32]. Information is scattered across platforms — reviews on Google Maps, event listings on Facebook, posts on Reddit, or short videos on TikTok — making it difficult to form a cohesive understanding. Increasingly, people tend to turn to peer-generated content [29, 32], especially short videos, for richer insights into local environments [4, 10]. These videos offer immersive, narrative-rich portrayals of place: walking tours, daily routines, or event vlogs that convey how people see, hear, and interact with their surroundings. CSCW scholars have long emphasized the importance of such multimodal and experiential cues in understanding place meaning [5, 6, 10, 18, 33]. However, the lack of systematic methods to retrieve such scattered and unstructured information limits its broader utility for spatial orientation and community integration.

To address this gap, we center our work on the framework of *localness*: how people experience, relate to, and become situated in specific places through cultural practices, social ties, and embodied routines [11, 13, 15]. Localness captures more than physical location: it reflects feelings of belonging, knowledge of community

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norms, familiarity with language or landmarks, and emotional attachment, etc. [11]. We aim to build computational systems that can capture these multidimensional aspects of place in accessible ways. Our approach introduces a graph-enhanced, retrieval-augmented generation framework for encoding localness from social videos. Compared to traditional RAG approaches that focus primarily on semantic similarity for retrieval, our graph-enhanced approach explicitly models the relationships among place-related information, enabling retrieval that considers both semantic relevance and structural connectivity within the place narrative [16, 23, 39].

We focus on TikTok as our primary data source due to its short-form, narrative-rich structure, its affordances for multimodal storytelling, and its widespread global adoption — particularly among youth and diverse communities [3, 8, 27, 41] — which enables access to vernacular place experiences often underrepresented in traditional data sources [4, 36, 37]. Specifically, starting with TikTok videos, we systematically construct a knowledge graph that captures the narrative and relational context of how places are depicted, and use this graph to guide LLMs in generating structured evidence-grounded localness attributes. This approach enables us to preserve both the sensory richness and the social fabric essential to place-based reasoning [1, 19, 21]. This study is guided by two research questions:

- **RQ1: How can multimodal social media content be systematically analyzed to construct place knowledge graphs?**
- **RQ2: Can graph-enhanced retrieval and prompt-guided LLMs reliably extract grounded, fine-grained localness attributes?**

Our contributions are threefold: (i) This study introduces a methodological framework for constructing place knowledge graphs from multimodal user-generated content; (ii) We develop a graph-enhanced retrieval-augmented generation approach for encoding localness as place representation; (iii) This study provides empirical insights of the strengths and limitations of the decoding social media content and encoding localness workflow. Ultimately, we aim to support the development of location-based services that help people understand, belong, and thrive in the places they call *home*.

## 2 Workflow of Capturing Localness from Multimodal Social Media Content

Our goal is to extract interpretable, context-rich representations of localness from TikTok videos. To do so, we propose a Graph-RAG framework that integrates multimodal feature extraction, structured knowledge representation, and LLM reasoning (Figure 1).

### 2.1 Multimodal Posts to Knowledge Graph

**2.1.1 Modality Alignment.** The first step is to convert multimodal TikTok posts, including video, audio, and text, into a unified text format to support downstream knowledge graph construction and retrieval. Inspired by prior studies that apply MLLMs for video analysis [24, 25], we first segment videos into keyframes, and use GPT-4o to generate a scene description for each keyframe. For fine-grained feature extraction, we detect objects, landmarks, and on-screen text in each keyframe and represent all outputs in text

form. We use Whisper to transcribe audios and segment them according to the corresponding time windows of each keyframe. Full modality alignment details are provided in Appendix A. Eventually, all multimodal TikTok content is extracted and converted into text, including post text, frame transcripts, detected landmarks, on-screen text, and scene descriptions.

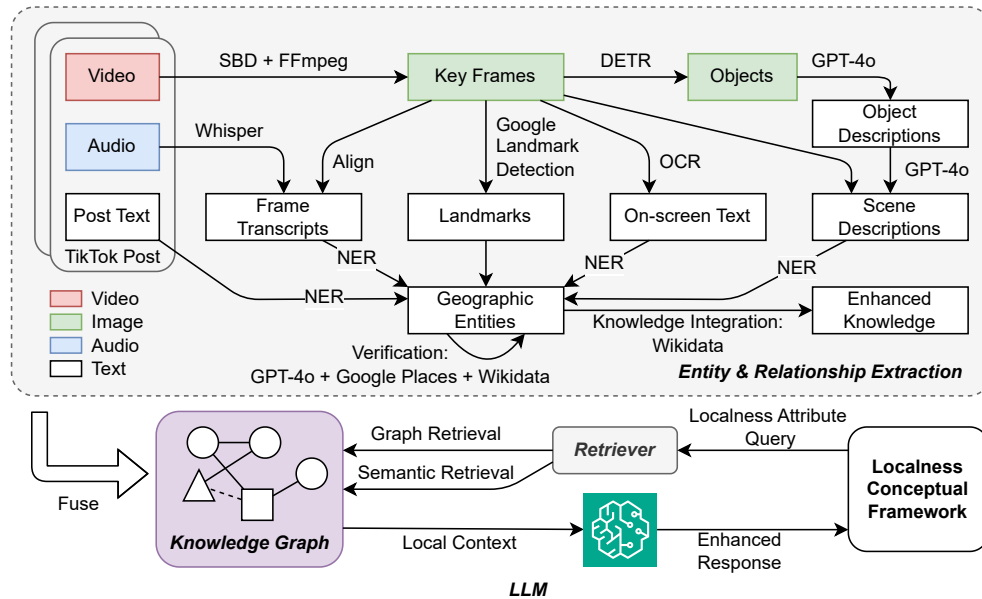
**2.1.2 Geographic Entity Recognition and Verification.** To associate the extracted elements with locations, we use spaCy’s NER to extract named locations and cultural entities, and GPT-4o to filter ambiguous mentions like “the Square” or “the Heights” with contextual reasoning. We verify the extracted locations to ensure they are clearly mapped to real locations. We use the Google Places API to map them to official names, formatted addresses, coordinates, and related metadata. We then query Wikidata for detailed descriptions and compute their semantic similarity with the corresponding frame transcripts and scene descriptions for cross-validation. If their cosine similarity exceeds a certain threshold, we consider the location verified. We reuse the vague mentions identified by GPT-4o by linking them to the verified place as local or contextual aliases. This preserves local language while grounding it in real-world geography.

**2.1.3 Fusing Entities and Relationships into Knowledge Graph.** The modality alignment and geographic entity recognition process naturally populates a set of entities and their relationships. For example, a geographic entity extracted from a scene description is linked to its corresponding keyframe. We construct the knowledge graph using NetworkX MultiDiGraph in PostgreSQL, where entities and relationships serve as nodes and edges. To enable semantic retrieval in addition to graph-based queries (Section 2.2.2), we also generate embeddings for all entities and relationships in this step.

## 2.2 Encoding Localness Attributes

**2.2.1 Localness Framework.** We introduce the graph-enhanced RAG framework for inferring *localness attributes* — fine-grained characteristics associated with specific places. Our methodology leverages the structured Localness Conceptual Framework from Gao et al. [11], which organizes localness into three interconnected domains: *Physical*, *Cognitive*, and *Relational*. Each domain consists of clearly defined dimensions and components. The *Relational* domain captures social connections and emotional bonds (e.g., community engagement, friendships). The *Physical* domain involves direct interaction with a place (e.g., long-term residence, community gardening). The *Cognitive* domain focuses on local knowledge and cultural understanding (e.g., knowing local history, navigating without assistance). This structured, hierarchical framework enables precise and intuitive annotation by illustrating how various aspects of localness manifest in different place.

**2.2.2 Hybrid Retrieval and Prompt Engineering.** We combine two key components: hybrid retrieval and framework-guided prompt engineering, to generate grounded and interpretable localness attributes. **Hybrid retrieval** integrates two complementary strategies to gather relevant evidence for a given location: (1) graph retrieval collects topologically adjacent nodes from the knowledge graph in up to 2 hops, (2) semantic retrieval identifies semantically similar nodes (e.g., other videos describing similar experiences)



### Figure 1: Workflow Visualization

with text embeddings. This ensures both relational and conceptual relevance in the retrieved evidence. For example, for a small neighborhood café, the retrieved context might include: transcripts where users describe the café as a “quiet local spot to study”; scene descriptions showing people working on laptops; spatial relationships linking the café to a nearby bookstore and park frequently mentioned in the same video; and audio cues such as regional speech patterns or ambient jazz music. **Prompt engineering** is structured around the Localness Conceptual Framework. This framework governs both the format of the prompt and the expected output structure. At inference time, we embed the retrieved evidence into a prompt template that instructs the LLM to extract localness attributes that are specific to the target location solely based on the provided context. The prompt includes detailed instructions to ensure quality control: the LLM must distinguish between direct observation and inference, cite concrete supporting evidence (e.g., quotes, timestamps, visual descriptions), and assign confidence levels. This hybrid strategy ensures that LLM outputs are not only semantically rich and relationally aware, but also evidentially grounded

## 2.3 Human Evaluation

To assess the effectiveness of our framework in capturing localness, we conducted human evaluation using the evidence retrieved by the Graph-RAG system. Two human coders independently annotated localness attributes for each target location, following the same schema used in LLM prompting. Prior to reconciliation, inter-rater reliability was assessed using Cohen’s kappa ( $\kappa = 0.72$ ), indicating substantial agreement between coders. They then reconciled their annotations through discussion to create a ground truth for evaluating the LLM’s output.

We compared LLM-generated attributes against this ground truth using three metrics: (1) *Recall* – the proportion of human-annotated attributes identified by the LLM; (2) *Precision* – the semantic similarity between LLM and human attributes; and (3) *Hallucination Rate*: – the proportion of LLM-generated attributes that are neither present in the human annotations nor directly supported by retrieved evidence. This metric offers a lens into overgeneration behaviors and grounding failures. To complement these metrics, we conducted a content analysis of the generated attributes, reasoning, and evidence to better understand the quality, potential, and limitations of the approach.

### 3 Evaluating Localness Attributes Inference

We selected 3 geotags corresponding to different U.S. cities and collected a dataset comprising 1,033 TikTok videos. From these videos, we extracted 823 verified location mentions using our multimodal entity verification approach. We randomly sampled 28 locations appearing in 45 videos and created ground truth annotations of their localness attributes. By comparing ground-truth with generated attributes, we found that the integration of graph-enhanced retrieval and structured prompting proved effective in extracting nuanced localness attributes from TikTok videos. By combining human evaluation metrics — *coverage*, *semantic similarity*, and *hallucination rate* — with grounded content analysis, we demonstrate the strengths and limitations of our framework in modeling place representation through social video data.

✓**The Workflow Achieves Complete Coverage Across All Localness Components.** This indicates that our graph-enhanced RAG approach is able to activate the complete spectrum of place attributes identified in human annotations. The extracted attributes

**Table 1: Domain components and evaluation metrics**

| Localness Component                 | Description                                                                                                 | Cov. | Halluc. | Sim.   | Items |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------|------|---------|--------|-------|
| <b>Localness Domain: Cognitive</b>  |                                                                                                             |      |         |        |       |
| Change Awareness                    | Recognition of how the place is evolving and developing over time.                                          | 1    | 0.3333  | 0.6667 | 12    |
| Geographic Familiarity              | Understanding spatial layout, recognizing landmarks, and knowing local boundaries and land features.        | 1    | 0.4103  | 0.5510 | 49    |
| Food Culture                        | Understanding and appreciation of the place's culinary traditions.                                          | 1    | 0.5500  | 0.4929 | 30    |
| Historical Knowledge                | Understanding the place's past and how it has evolved over time.                                            | 1    | 0.1111  | 0.7787 | 18    |
| Ecological Understanding            | Awareness of seasonal patterns, environmental cycles, local watersheds, and environmental challenges.       | 1    | 0.1667  | 0.7287 | 17    |
| Local Recommendations               | Having detailed knowledge about local establishments and services.                                          | 1    | 0.0833  | 0.8123 | 25    |
| Natural Environment                 | Knowledge of local flora and fauna, environmental literacy, and understanding natural systems and patterns. | 1    | 0.0000  | 0.8134 | 8     |
| Language and Dialect                | Familiarity with the unique linguistic features of the place.                                               | 1    | 0.3810  | 0.6773 | 32    |
| Local Customs and Norms             | Understanding unwritten social rules and behaviors specific to the place area.                              | 1    | 0.1538  | 0.7290 | 25    |
| <b>Localness Domain: Physical</b>   |                                                                                                             |      |         |        |       |
| Formative Years                     | Having spent developmental years in the place, shaping one's worldview.                                     | 1    | 0.5000  | 0.5000 | 22    |
| Geographic Familiarity              | Practical navigation skills and spatial orientation in the local area.                                      | 1    | 0.4103  | 0.5510 | 49    |
| Ecological Understanding            | Physical connection and embodied familiarity with local ecosystems.                                         | 1    | 0.1427  | 0.7926 | 17    |
| Environment Experience              | Reflects a personal, physical connection to natural spaces and local landscapes.                            | 1    | 0.2000  | 0.7000 | 12    |
| <b>Localness Domain: Relational</b> |                                                                                                             |      |         |        |       |
| Civic Engagement                    | Participating in local governance and political processes in the area.                                      | 1    | 0.5769  | 0.4615 | 35    |
| Sense of Belonging                  | Feeling comfortable and accepted within the place.                                                          | 1    | 0.1538  | 0.7900 | 37    |
| Active Participation                | Direct involvement in community activities and initiatives in the place.                                    | 1    | 0.1538  | 0.7409 | 23    |
| Feeling of Home                     | Emotional attachment to the place that creates security and comfort.                                        | 1    | 0.1154  | 0.7949 | 47    |
| Community Investment                | Demonstrating care and commitment to the wellbeing of the area.                                             | 1    | 0.0417  | 0.8295 | 21    |
| Identity Connection                 | Incorporating the place into one's self-concept and identity.                                               | 1    | 0.1923  | 0.7424 | 32    |

Note: Cov. = Coverage, Halluc. = Hallucination rate, Sim. = Similarity, Items = Total number of evaluated instances.

demonstrate the system's ability to represent place beyond physical descriptions alone, capturing the socio-cultural dimensions that define how communities experience places.

✓ **Physically Observable Attributes Are Captured with High Precision and Minimal Hallucination.** For example, *Natural Environment* scored perfectly with 0% hallucination and a high similarity score of 81.3%, suggesting that environmental features are easier to ground using visual and narrative content. This is consistent with observations from locations such as *springs\_recreation\_area\_A*:

"The *springs\_recreation\_area\_A* features a diverse ecological system with winter landscapes that include snow-covered paths and a frozen lake."

The attribute is grounded in direct visual evidence from videos showing snowy landscapes with people walking on frozen lakes.

✓ **Emotional and Relational Dimensions Are Effectively Extracted from Social Media Cues.** Attributes tied to emotional connection, such as *Feeling of Home*, *Sense of Belonging*, and *Identity Connection*, demonstrated both high similarity scores (> 74%) and low hallucination rates (11.5%, 15.4%, and 19.2%, respectively). This suggests that emotional cues, especially when expressed through hashtags, enthusiastic captions, and visual event participation scenarios, are particularly amenable to LLM-based inference. In the case of *block\_party\_B*, the model accurately captured emotional resonance and collective rituals through phrases like:

"The location fosters a sense of community among its residents, as evidenced by social interactions during events."

This attribute reflects how specific social rituals like the *block\_party\_B*, serve as anchors for community identity and belonging, both detectable through repeated social media themes.

✓ **Temporal and Relational Context Provides Foundation for Advanced Place Reasoning.** Certain components, like *relational context* and *temporal dimensions* stood out for their balance of low hallucination and high similarity. *Community Investment* (hallucination: 4.2%, similarity: 82.9%), *Historical Knowledge* (11.1%, 77.9%), and *Local Recommendations* (8.3%, 81.2%) exemplify categories where local knowledge is often stated directly or implied with strong supporting evidence. For example, in the restaurant context:

"*restaurant\_C* was a predecessor to the current *restaurant\_D*, indicating its historical significance in the area."

The system not only identifies individual places but situates them within historical narratives and evolving relationships. Similarly, the system's sensitivity to temporal dimensions is evident in its ability to represent how places transform across seasons:

"The *natural\_landscape\_E* experiences significant seasonal changes, particularly in winter when the area is popular for outdoor activities like walking on the ice and ice fishing."

These attributes capture the dynamic nature of place experience across time — something traditional place representations often miss. The relatively strong performance metrics for temporally-oriented categories validate the graph-enhanced approach's effectiveness in capturing these crucial dimensions of place.

? **The Framework Overgeneralizes Cultural Patterns from Sparse Cues.** Despite prompt-level instructions to avoid using prior knowledge, the system frequently overgeneralizes cultural attributes from minimal evidence. *Food Culture* and *Language and Dialect* showed high hallucination rates (55% and 38.1%, respectively), often extrapolating broad cultural claims from isolated mentions. For example, a single dish — cheese curds — led to the attribution:

“This place is known for its dairy-based cuisine.”

These culturally plausible yet unsupported inferences suggest the LLM fills contextual gaps using learned priors when retrieved content is sparse. This limitation is especially salient for culturally nuanced components requiring deeper, grounded understanding of local practices.

**? The System Hallucinates Social and Civic Attributes in the Absence of Evidence.** *Civic Engagement*, in particular, exhibited the highest hallucination rate (57.7%) and the lowest semantic similarity score (46.2%). Unlike cultural overgeneralization, which arises from minimal evidence, these hallucinations occur in the complete absence of supporting data. For instance, in videos from a *student\_neighborhood\_F*, the model inferred community activism and political engagement, despite no visual or textual indicators of such behavior. These hallucinations occurred even though prompts explicitly instructed the LLM to skip attributes without grounded evidence. This suggests a deeper limitation: the model defaults to filling gaps with plausible social narratives, especially for dimensions of localness that are abstract or difficult to observe visually.

## 4 Discussion

### 4.1 Graph-RAG Successfully Captures Rich Place Representations While Enabling Transparent Reasoning

**4.1.1 Graph-RAG enables structured, situated representations of localness.** Our findings confirm that the Graph-RAG framework is a powerful approach for generating comprehensive, fine-grained representations of place. By structuring TikTok-derived information into a place knowledge graph, the system achieved full coverage across all localness components we examined, including physical, cognitive, and relational domains of place. In particular, it excelled in categories reflecting lived experience, such as identity, belonging, and community connection, which are often hard to quantify. This suggests our framework successfully extracts the emotional, cultural, and social layers of locality that conventional location-based services tend to overlook. The ability to surface insider perspectives (e.g. local expressions, traditions, or norms) has practical implications for systems supporting newcomers and community members [18, 33]. In short, graph-guided retrieval allows an LLM to paint a more human-centric portrait of place, preserving the contextual nuances essential to place-based reasoning [8].

**4.1.2 Graph grounding improves transparency and supports community trust.** Unlike end-to-end LLMs, our graph-enhanced pipeline allows users to trace where specific inferences originated (e.g., which videos or captions supported an emotional attribute). Interpretability of model outputs is critical in civic and community-facing systems [35]. By linking outputs to evidence nodes, the system fosters transparency and enables community members to validate, contest, or augment representations of place. Future work could incorporate interactive maps or visualizations to make this transparency user-facing, similar to place graph tools explored in urban informatics [6].

**4.1.3 Component analysis can isolate contributions and failure points.** Our current evaluation confirms the effectiveness of the full pipeline,

but does not disentangle which components (e.g., graph retrieval, semantic similarity, visual tagging) contribute most to success. Ablation studies, e.g., comparing outputs with vs. without graph retrieval, could clarify the added value of structured memory. Prior work in RAG architectures shows that relational retrieval can significantly reduce hallucinations in knowledge-poor settings [22]. Likewise, a modality-level error breakdown (e.g., visual vs. auditory sources) could identify where most grounding failures originate and guide improvements.

### 4.2 Technical Limitations Reveal Systematic Challenges in Grounding and Bias Mitigation

**4.2.1 Hallucinations reflect sparsity and overgeneralization in social video.** Despite strong coverage, our results revealed notable limitations in cultural and social reasoning. High hallucination rates in categories like *Civic Engagement* and *Food Culture* underscore the challenge of generating grounded localness attributes from sparse or implicit evidence. As Quercia et al. [33] warned, even multi-modal data can yield impoverished representations when deeper context is absent. Although our prompts instructed the model to skip unsupported claims, it frequently defaulted to plausible but ungrounded inferences, especially for abstract or institutional attributes. Future work should target two key improvements. First, better cross-referencing of evidence across multiple videos and sources can improve grounding, particularly for abstract or underrepresented dimensions of localness. Second, refining prompt strategies may help reduce hallucinations in socially sensitive categories, such as political participation or community dynamics. Expanding to include video data from multiple platforms (e.g., Instagram, YouTube) will also address the challenge of fragmented representations across digital ecosystems, offering a more holistic view of place experience.

**4.2.2 Biases and upstream errors may compound through the pipeline.** Like other social media-based systems, our model is exposed to selection biases in content creation. TikTok content skews toward younger, visually expressive users, and trends often over represent commercial or aesthetic places over utilitarian or marginalized spaces [3, 27, 38]. These biases risk reinforcing dominant narratives while excluding overlooked communities. Moreover, minor upstream errors, like OCR misreads or object misclassification, may propagate through the graph and amplify in final inferences. Such cascading failure is a known risk in long pipelines. Addressing this in future works requires diverse input sampling, debiasing modules, and post-hoc verification using population-level data (e.g., census records or local surveys).

### 4.3 Generalizability and Ethical Responsibility

**4.3.1 Cultural generalization requires adaptation and multilingual grounding.** Our study focused on English-language TikTok data from U.S. cities, but notions of localness are culturally situated and may not transfer directly. Place attachment and civic participation, for instance, manifest differently across geographies and governance models. Additionally, digital footprints are unequally distributed: many communities are “data poor” or invisible online

[17, 26, 28]. To generalize globally, future works will require multi-lingual support, culture-aware tuning, and collaborative refinement of the localness taxonomy through cross-regional studies.

**4.3.2 Ethical safeguards and participatory input are essential for responsible use.** Even publicly shared content requires careful ethical handling. TikTok videos often feature identifiable people, routines, or communities, raising concerns about privacy and misrepresentation. We adopted basic anonymization and sensitive content filtering, but future work should integrate participatory annotation, consent-aware data governance, and adversarial red teaming [9, 31]. Drawing from contextual integrity theory [30], we argue that content must be interpreted and used in ways aligned with its original audience expectations. Working with community moderators and documenting dataset construction (e.g., via datasheets) can enhance both ethical rigor and public trust [12, 14].

#### 4.4 Beyond Place Representation: A Generalizable Approach to Fragmented, Multimodal Content

More broadly, this workflow offers a generalizable method for synthesizing meaning from fragmented, multimodal social content. While developed for place representation, the graph-enhanced RAG approach can extend to other CSCW domains, such as crisis informatics, online health communities, or migration studies, where understanding depends on piecing together narrative, visual, and audio signals across noisy media [25]. As social content continues to diversify in form and platform, methods that integrate structure and grounding will be increasingly important for making sense of complex human experiences.

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## A Modality Alignment Details

**Visual Keyframe Extraction:** We apply Shot Boundary Detection (SBD) using PySceneDetect<sup>1</sup> to identify major visual transitions. Representative frames are then extracted at these boundaries using FFmpeg, ensuring essential visual content is retained for further analysis. We recorded each Keyframe’s timestamp for the following key frame time-aligned transcript segment extraction.

**On-screen Text Extraction:** We apply OCR to extracted keyframes using PyTesseract<sup>2</sup>, capturing textual elements embedded in visuals — such as signs, hashtags, and subtitles—which often carry crucial contextual information.

**Visual Object Detection:** Using DETR (DEtection TRansformer) [2], we identify and localize salient objects within each video keyframe. DETR’s transformer-based architecture enables robust detection even in cluttered or dynamic environments common in user-generated videos. Following object detection, we generate descriptive scene descriptions using the GPT-4o vision-language model. This model ingests both raw keyframes and previously detected objects, producing semantic captions that help contextualize the scene beyond surface-level object recognition.

**Landmark Recognition:** To resolve place-specific semantics, we apply the Google Landmark Recognition API<sup>3</sup> to identify notable geographical or cultural landmarks present in keyframes. Detected locations are further verified by comparing them with entities extracted from other data, the detailed verification process is in Section 2.1.2. To further enhance semantic depth, we integrate structured knowledge from Wikidata. These resources provide structured metadata (e.g., location type, cultural tags, region hierarchy), enriching both the visual and textual representations with authoritative external information.

**Audio Transcription:** We extract textual information from both spoken and embedded sources. Audio segments are transcribed

using OpenAI’s Whisper model [34]. Whisper is particularly well-suited due to its multilingual capabilities and robustness to real-world noise, a common feature in user-generated TikTok videos. We also extract both semantic and paralinguistic features. We employ pre-trained PANNs [20] and YAMNet models to tag environmental sounds and audio events, such as ambient noise, music genres, or human activity. Additionally, SpeechBrain’s ECAPA-TDNN model [7] is used to generate speaker embeddings that reflect accents, voice timbre, and other sociolinguistic cues potentially indicative of local or regional identity.

**Temporal Alignment:** Temporal patterns are incorporated through ByteTrack [40], which tracks detected objects across frames. This adds continuity to spatial dynamics and captures context-specific activities—such as people walking in front of a monument or traffic near a city square—which can provide additional cues about location types and usage. Using timestamp information, we synchronize the extracted keyframes, transcribed text, OCR outputs, and audio features to form a cohesive, time-aligned representation of each video.

<sup>1</sup><https://github.com/Breakthrough/PySceneDetect>

<sup>2</sup><https://github.com/madmaze/pytesseract>

<sup>3</sup><https://developers.google.com/maps/documentation/landmark>